



Simultaneous classification of hand gestures and elbow angles using sEMG signals: a comparison of machine learning performances

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Abstract

This study evaluated the preliminary feasibility of simultaneously classifying hand gestures and corresponding elbow angles using only surface electromyography (sEMG) signals and lightweight machine learning models. Conventional human–machine interface (HMI) systems often separate gesture recognition from angle estimation or require additional sensors, such as inertial measurement units, thereby increasing hardware complexity and cost. sEMG signals were collected from eight participants using a five-channel configuration while performing four hand gestures at two discrete elbow-angle conditions, yielding a total of eight classes under controlled experimental settings. The signals were preprocessed, segmented using a sliding-window approach, and converted into five standard time-domain features, which were used to train support vector machine (SVM), random forest (RF), and extreme gradient boosting (XGBoost) classifiers. Among the evaluated models, the RF classifier achieved the highest mean accuracy, reaching 96.72% ($\pm 0.58\%$) across five-fold cross-validation. Misclassification analysis showed that most errors occurred between elbow-angle conditions within the same gesture rather than between different gestures. These findings demonstrate the preliminary feasibility of simultaneously classifying hand gestures and elbow-angle conditions using only sEMG signals and lightweight machine learning models under controlled experimental conditions. This sensor-efficient framework may support the future development of simplified multidimensional motion-intent recognition systems for applications in prosthetic control, rehabilitation robotics, and other HMI systems.

Keywords Surface electromyography · Hand gesture recognition · Elbow angle estimation · Simultaneous classification · Machine learning

1 Introduction

The rapid development of artificial intelligence and robotics has increased the demand for intuitive and reliable human–machine interface (HMI) technologies. Accurately decoding a user’s motor intent is a critical challenge in applications such as myoelectric prosthetic hands, rehabilitation

robotics, and virtual reality controllers [1]. Among various non-invasive bio-signals used for this purpose, surface electromyography (sEMG) is widely adopted because this technique records muscle activation patterns from the skin surface and can reflect user intention in real time [2]. These signal characteristics render sEMG a promising modality for controlling assistive and interactive systems [3].

Traditional research on sEMG-based motion recognition typically focuses on identifying a single aspect of user intent, such as hand gesture classification [4] or elbow angle estimation [5]. Although effective in isolated tasks, this separation limits the ability to interpret more complex multi-degree-of-freedom movements that are characteristic of daily activities. To address this challenge, prior studies have explored the simultaneous estimation of gestures and joint angles using sEMG combined with additional sensors, such as inertial measurement units (IMUs) [6]. Although multi-sensor fusion improves estimation accuracy, system cost

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increases, data synchronization issues occur, and hardware complexity is introduced, which may reduce practicality in wearable or clinical applications [7].

Recent studies have demonstrated that elbow joint angles can be classified using only sEMG signals, demonstrating the feasibility of applying a single-sensor approach for kinematic estimation [8]. However, these studies were limited to single-degree-of-freedom movements and did not address the broad objective of simultaneously decoding both hand gestures and joint angles. Moreover, deep learning models, such as convolutional and recurrent neural networks (CNNs and RNNs, respectively), have garnered attention for their ability to extract features directly from raw sEMG signals [9, 10]. However, deep learning methods generally require large datasets, high computational power, and considerably long training times, rendering them less suitable for early-stage explorations or resource-constrained environments, where interpretability and computational efficiency are essential.

To advance the feasibility of simplified, sensor-efficient recognition of motion intent, this study extends previous studies by systematically comparing the performances of three widely used machine learning models, namely, random forest (RF), support vector machine (SVM), and extreme gradient boosting (XGBoost), to simultaneously classify multi-degree-of-freedom hand gestures and elbow angles using only a single sEMG array. Unlike prior studies that employed additional inertial sensors or considered one type of motion, this study evaluates whether lightweight machine learning algorithms, in combination with standard time-domain features, can effectively classify multi-class motion intent from sEMG alone under controlled experimental conditions.

The primary contribution of this study is the preliminary experimental validation that computationally efficient machine learning models can simultaneously classify both gesture type and elbow-angle conditions under controlled experimental settings without requiring additional sensors or deep learning architectures. This study provides foundational evidence supporting the potential application of simplified hardware configurations and low-cost systems for future HMI-related research and assistive technologies, including prosthetic control and rehabilitation systems.

2 Materials and methods

2.1 Participants

Eight healthy participants (five males and three females; mean age 23 ± 2 years) with no history of neuromuscular disorders were recruited for this study. All participants were

right-handed. Although the cohort size was limited to eight individuals, multiple motion trials were collected from each participant, resulting in a total of 1920 labeled segments for machine learning analysis under controlled experimental conditions. While the dataset size supported within-subject classification evaluation, the limited number of participants may restrict subject-independent generalizability. All procedures were conducted in accordance with the Declaration of Helsinki and approved by the Institutional Review Board of Kyungpook National University Hospital (KNUH IRB 2025-08-024-003). Written informed consent was obtained from all participants.

2.2 Data acquisition and experimental protocol

A five-channel sEMG system (IX-BIO8, iWorx) was used to record muscle activity at a sampling rate of 1 kHz (Fig. 1a and c). Disposable Ag/AgCl electrodes were placed in a bipolar configuration on the dominant forearm and upper arm. The five recording channels corresponded to the extensor carpi radialis (Ch 1), extensor digitorum (Ch 2), flexor digitorum superficialis (Ch 3), biceps brachii (Ch 4), and triceps brachii (Ch 5). A ground electrode was attached near the radial styloid process (Fig. 2). Electrode locations were determined based on identifiable anatomical landmarks and corresponding muscle regions, and the placement procedure was applied consistently across all participants to minimize inter-subject variability.

The participants were seated with their arms resting on a table equipped with a custom angle-measurement plate to maintain consistent elbow-angle positions (Fig. 1b). The experiment involved eight classes, each defined by one of four hand gestures, namely, fist, open hand, pronation, and supination, performed at two elbow angles (30° and 60°) (Fig. 3). These movements were selected because they represent fundamental and frequently encountered actions in daily life and provide a controlled experimental baseline for evaluating simultaneous gesture–angle classification.

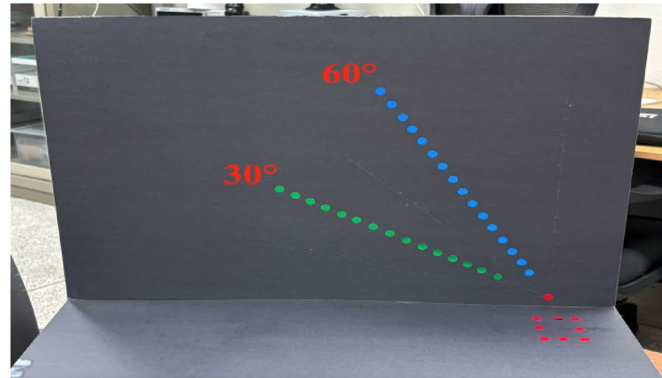
In addition, four static gestures and two discrete angles were intentionally selected to minimize the number of confounding variables and maintain consistent muscle activation patterns across repetitions. Using static tasks ensured a controlled experimental environment for assessing the preliminary feasibility of simultaneous classification prior to investigating dynamic or more complex movements in future work.

The participants followed a metronome set at 60 beats/min. Each trial lasted 5 s, consisting of 2 s of rest, 1 s of motion execution, and 2 s of rest. Each class was repeated 30 times, yielding 2 min and 30 s of sEMG data per class and a total of 240 motion trials corresponding to each participant.

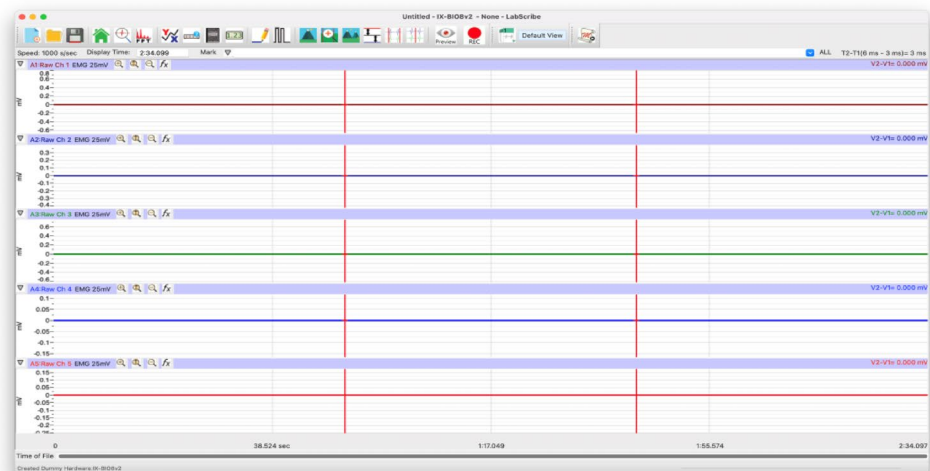
Fig. 1 Experiment equipment. **a** IX-BIO8 for recording sEMG signals using Chs 1, 2, 3, 4, and 5 and ground. **b** Angle measurement plate for the elbow joint. **c** Software used for data collection



(a)



(b)



(c)

2.3 Signal processing and feature extraction

The raw sEMG signals were pre-processed to remove noise. A 60 Hz notch filter was employed to eliminate power-line interference, followed by a 20–450 Hz fourth-order Butterworth bandpass filter to retain the primary frequency range of the sEMG signal.

The filtered 2-s (2000-sample) segments were subsequently used for feature extraction by employing the sliding window technique. A 200-ms (200-sample) analysis window was applied to each segment with a 50% overlap, causing an increment (step size) of 100 ms (100 samples). From each 200-ms window, five standard time-domain features were extracted, namely, root mean square (RMS), mean absolute value (MAV), wavelength (WL), zero crossings

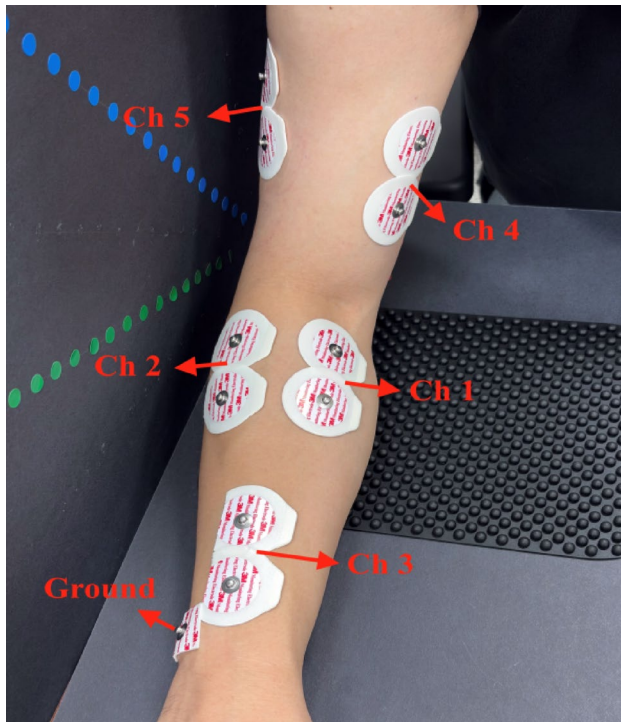


Fig. 2 sEMG electrode patch attachment location (extensor carpi radialis (Ch 1), extensor digitorum (Ch 2), flexor digitorum superficialis (Ch 3), biceps brachii (Ch 4), and triceps brachii (Ch 5), radial styloid process (ground))

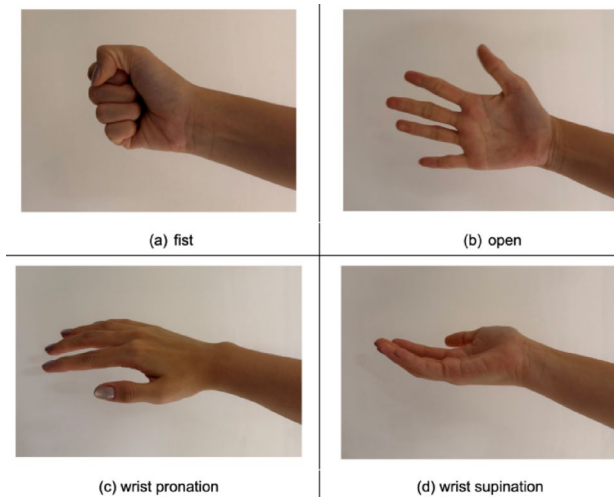


Fig. 3 Four hand gestures used in the experiment: **a** fist, **b** open, **c** wrist pronation, and **d** wrist supination

(ZC), and slope sign changes (SSC). This process yielded a feature vector of 25 dimensions (5 features $\times$ 5 channels) for each window. Each 2-s segment yielded 19 such feature vectors (windows).

The five features extracted, where N is the number of samples in the window ($N=200$) and x_t is the t th sample, are as follows:

RMS: This feature represents the square root of the mean of the squared signal values, which is related to the signal power and amplitude.

$$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2} \quad (1)$$

MAV: This feature indicates the average absolute amplitude of the sEMG signal, a common and effective measure of muscle contraction level.

$$MAV = \frac{1}{N} \sum_{i=1}^N |x_i| \quad (2)$$

WL: This is the cumulative length of the signal waveform, providing information regarding signal complexity, amplitude, and frequency.

$$WL = \sum_{i=2}^N |x_i - x_{i-1}| \quad (3)$$

ZC: This feature counts the number of times the signal crosses the zero-amplitude axis, providing an estimation of the signal's frequency content. To minimize the influence of noise, a small threshold (th) of 0.01 was applied, and ZC was calculated as shown in Eq. (4).

$$ZC = \sum_{i=2}^N f((x_{i-1} \times x_i) < 0 \text{ and } |x_i - x_{i-1}| \geq th) \quad (4)$$

SSC: This feature counts the number of times the slope of the signal changes sign (from positive to negative or vice versa), additionally reflecting frequency information.

$$SSC = \sum_{i=2}^{N-1} f((x_i - x_{i-1}) \times (x_i - x_{i+1}) > 0) \quad (5)$$

Only time-domain features were used in this study. Although frequency-domain features (e.g., mean and median frequencies) and time-frequency features (e.g., wavelet coefficients) are widely used in sEMG research, applying them generally introduces additional computational complexity. The objective of this study was to evaluate the preliminary feasibility of efficient and lightweight feature extraction for multi-class sEMG classification under controlled experimental conditions. Therefore, time-domain features were selected, as they can be computed rapidly, are robust, and have been widely applied in EMG-based motion recognition studies.

Prior studies have demonstrated that time-domain features provide competitive accuracy for upper-limb motion classification while maintaining high computational efficiency, rendering them suitable for lightweight and computationally efficient EMG classification frameworks with potential future applicability to real-time or embedded systems.

2.4 Machine learning models

The extracted feature vectors were standardized using a StandardScaler (z-score normalization). Subsequently, the dataset was used to train and evaluate three machine learning models. Specifically, stratified K-fold cross-validation with randomized trial-wise splitting was used to preserve class balance across folds and evaluate classification performance. Because feature vectors were shuffled prior to splitting, samples from the same participants could appear across different folds.

SVM: The SVM model was implemented using the Scikit-learn library. To effectively classify non-linear sEMG features, the kernel was set to 'rbf' (radial basis function). To conduct the eight-class multi-class classification, the decision_function_shape was set to 'ovr' (one-vs-rest). The default values of regularization parameter C and the kernel coefficient, gamma, were retained (1.0 and 'scale', respectively); these values are commonly reported to provide stable baseline performance across various datasets [11]. The RBF kernel was selected because in prior sEMG studies, strong performance corresponding to nonlinear and subject-dependent muscle activation patterns has been consistently reported. The default C and kernel coefficient gamma (γ) in Scikit-learn are frequently used as reliable baseline settings in biomedical machine learning studies without requiring extensive hyperparameter tuning.

RF: The RF model was implemented using Scikit-learn. The key parameter, n_estimators (number of decision trees), was set to 100, balancing performance and computational efficiency. This value was selected to provide a sufficient ensemble size for reducing variance without incurring high computational cost. To ensure reproducibility, random_state was fixed to 42. Moreover, n_jobs was set to -1 to utilize all CPU cores for rapid training. The values of the other parameters, such as the split quality criterion and maximum depth, were retained as their default values [12]. The choice of 100 estimators was based on previous EMG classification research, in which tree counts between 50 and 200 revealed minimal performance differences while considerably affecting computation time. Using 100 trees provided an effective

balance between accuracy, model stability, and computational efficiency, rendering this a practical configuration for performance comparison.

XGBoost: Unlike the other models, the key hyperparameters applied in XGBoost were configured based on preliminary tuning to improve classification performance. The objective was set to 'multi: softprob' to predict the probability corresponding to each class. The number of boosting rounds (n_estimators) and the tree complexity (max_depth) were set to 400 and 6, respectively. Moreover, a low learning_rate of 0.05 was used for stable learning. To prevent overfitting, subsample was set to 0.9, colsample_bytree to 0.8, and reg_lambda (L2 regularization) to 1.0. Additionally, the random_state was fixed to 42 for reproducibility, and the tree_method was set to 'hist' for fast, CPU-based training [13]. These hyperparameters were selected based on preliminary tuning and recommendations from prior EMG-related machine learning studies, which identified that medium-depth trees and conservative learning rates prevent overfitting of physiological signals. Furthermore, the subsampling and regularization parameters were selected to reduce sensitivity to noise and subject-specific variations.

2.5 Statistical analysis and evaluation

The performances of the models were evaluated using two validation approaches, namely, standard hold-out validation (80% training, 20% testing) and five-fold stratified cross-validation with randomized trial-wise splitting. Specifically, feature vectors were randomly shuffled prior to splitting, and stratification was applied to preserve the class distribution within each fold. The detailed data compositions used for training, validation, and testing are summarized in Table 1. Because the dataset splitting was performed at the trial level after random shuffling, samples from the same participants could appear across different training and testing folds. In addition, per-subject classification accuracies were calculated from the predictions obtained during the five-fold cross-validation to evaluate inter-individual variability in classification performance.

The classification performance was evaluated using four standard metrics, namely, accuracy, precision, recall, and F1-score, derived from the true positive (TP), false positive (FP), true negative (TN), and false negative (FN) counts [14].

These metrics were computed according to the following equations:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (6)$$

Table 1 Composition of data used for training and testing

	Number of data
Train	1536
Test	384

$$\text{Precision} = \frac{TP}{TP + FP} \quad (7)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

$$F_1\text{-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

where TP , FP , TN , and FN represent the numbers of true positive, false positive, true negative, and false negative cases, respectively.

Moreover, confusion matrices were generated to analyze misclassification patterns between classes in detail [14]. The final performance of each model was reported as the mean and standard deviation ($\pm$ SD) of the classification accuracy obtained from five-fold cross-validation [15]. In addition, paired t-tests were conducted using the cross-validation accuracies obtained from each fold to statistically compare classifier performance.

3 Results

The performances of the three machine learning models were systematically evaluated using hold-out validation and five-fold stratified cross-validation.

3.1 Hold-out validation performance

Hold-out validation revealed that all three models achieved high classification accuracy, exceeding 95%. Among the evaluated models, the RF model achieved the highest classification accuracy (96.88%) and F1-score (0.9687) under the evaluated experimental conditions. The SVM and XGBoost models also demonstrated high performance, achieving accuracies of 96.35% and 95.57%, respectively. The detailed performance metrics obtained from each model, including precision, recall, and F1-score, are listed in Table 2. Pairwise paired t-tests conducted using the five-fold cross-validation accuracies did not reveal statistically significant differences among the evaluated classifiers (RF vs. SVM: $p=0.1193$; RF vs. XGBoost: $p=0.1609$; SVM vs. XGBoost: $p=0.3739$).

3.2 Cross-validation performance and stability

To assess the generalizability and stability of the models, five-fold stratified cross-validation was performed. The results listed in Table 3 demonstrate consistent performance across different folds. Among the evaluated models, the RF model achieved the highest mean accuracy of 96.72%

Table 2 Performance comparison of the RF, SVM, and XGBoost models using hold-out validation

Model	Accuracy	Precision	Recall	F1-score
RF	0.9688	0.9695	0.9688	0.9687
SVM	0.9635	0.9641	0.9635	0.9635
XGBoost	0.9557	0.9567	0.9557	0.9559

Table 3 Five-fold stratified cross-validation results obtained from the RF, SVM, and XGBoost models

Model	Accuracy
RF	96.72 $\pm$ 0.58%
SVM	95.83 $\pm$ 0.79%
XGBoost	96.15 $\pm$ 0.56%

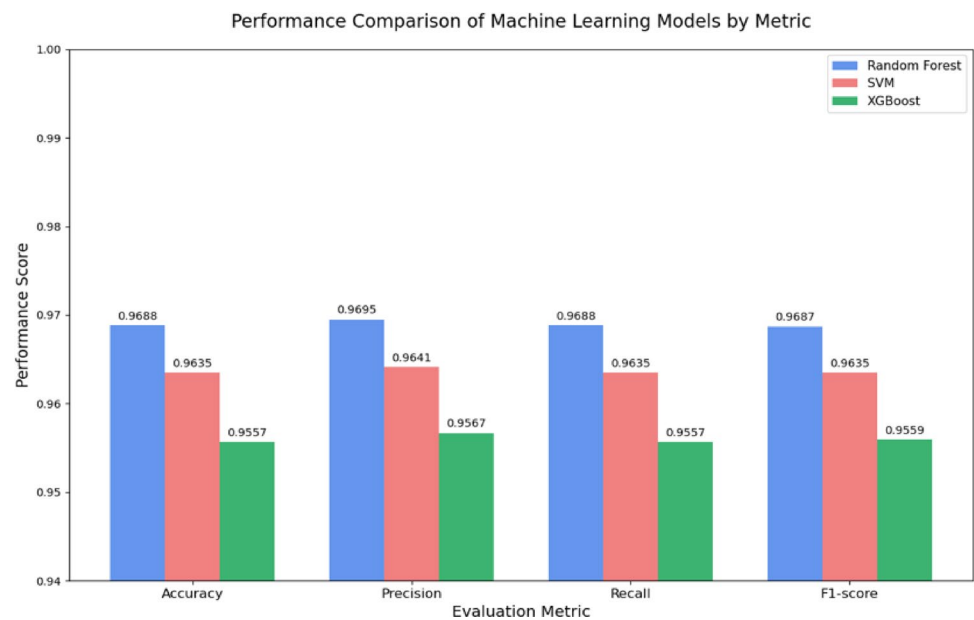
($\pm 0.58\%$), indicating stable performance across the cross-validation folds. In addition, the XGBoost and SVM models also demonstrated strong performance, with mean accuracies of 96.15% ($\pm 0.56\%$) and 95.83% ($\pm 0.79\%$), respectively. The low standard deviations obtained from all models (all < 0.008 or 0.8%) suggest that the proposed methodology was consistent and not heavily dependent on a specific data split (Fig. 4).

3.3 Per-subject classification performance

To further examine inter-individual variability, per-subject classification accuracies were calculated under the five-fold trial-wise cross-validation framework (Table 4). The Random Forest classifier achieved accuracies ranging from 94.2% to 100.0%, SVM from 87.5% to 100.0%, and XGBoost from 94.6% to 100.0% across the eight participants. These results indicate consistently high classification performance across all participants under the adopted trial-wise validation framework.

3.4 Computational cost analysis

To further evaluate the practical applicability of the evaluated machine learning models, computational cost was compared in terms of training time, inference time per sample, and saved model size. The results are summarized in Table 4. RF and SVM demonstrated relatively short training times (< 0.4 s), whereas XGBoost required substantially longer training time because of the boosted ensemble structure and tuned hyperparameters. However, all evaluated models maintained low inference times (< 0.4 ms/sample) and relatively compact model sizes (< 5 MB), supporting the feasibility of lightweight sEMG-based classification under controlled experimental conditions (Table 5).

Fig. 4 Performance comparison of the evaluated machine learning models using hold-out validation**Table 4** Per-subject classification performance under five-fold trial-wise cross-validation

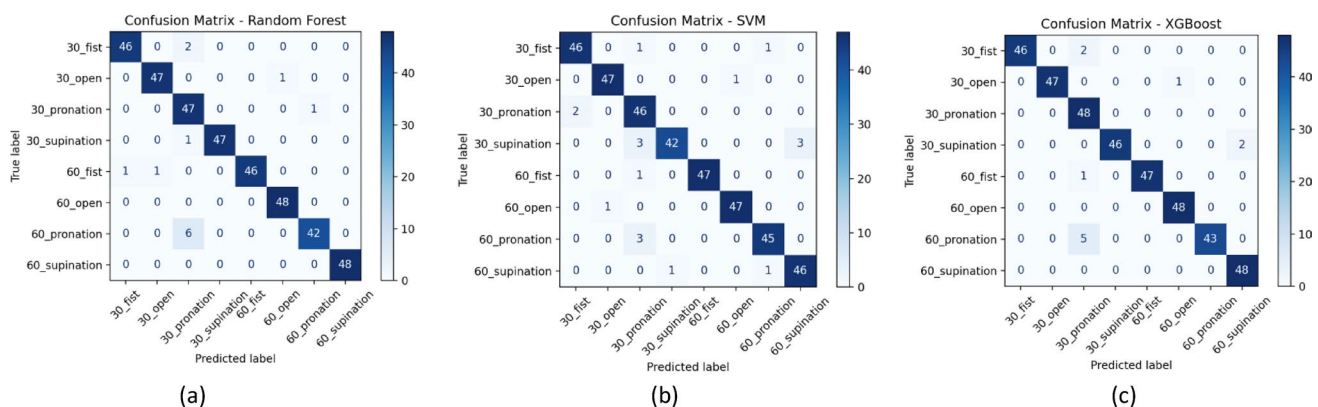
Subject	RF	SVM	XGB
GYJ	97.5	96.3	97.9
KDH	95.8	95.4	94.6
KMS	100	100	100
KYH	98.3	97.1	98.3
LDH	94.2	95.0	96.3
LSJ	94.2	87.5	95.4
LYT	97.1	97.9	95.4
SYJ	97.9	97.9	95.8

Table 5 Computational cost comparison of the evaluated models

Model	Training time (s)	Inference time per sample (ms)	Model size (MB)
RF	0.313	0.036	4.31
SVM	0.334	0.382	4.36
XGBoost	19.846	0.031	3.59

3.5 Misclassification analysis

To analyze error patterns in detail, confusion matrices were generated for all three models (Fig. 5). Similar misclassification patterns were observed across all models, with relatively few errors occurring between different gestures (e.g., fist vs. open). Most misclassifications occurred within the same gesture but between the two elbow angles. For instance, 30° fist was occasionally misclassified as 60° fist but rarely as 30° open hand. These findings suggest that the extracted time-domain features effectively discriminate among different hand gestures, whereas distinguishing subtle muscle activation differences between the 30° and 60° elbow angles remains relatively more challenging.

**Fig. 5** Confusion matrices for the eight-class classification. **a** RF, **b** SVM, **c** XGBoost

4 Discussion and conclusion

The primary objective of this study was to compare the performances of multiple machine learning models in terms of simultaneously classifying complex hand gestures and corresponding elbow angles using only sEMG signals, without employing any auxiliary sensors. The obtained results revealed that all three models, namely, RF, SVM, and XGBoost, achieved mean accuracies exceeding 95%, confirming that a single sEMG array can effectively capture multi-dimensional motion intent without requiring additional sensors, such as IMUs [6, 7]. Among these models, the RF classifier exhibited the highest and most stable cross-validation accuracy ($96.72\% \pm 0.58\%$) under the evaluated experimental conditions, although paired statistical comparisons did not reveal statistically significant differences among classifiers.

Class-wise performance analysis revealed consistent patterns across models. Gestures involving large and distinct muscle activations, such as an open hand gesture, yielded high classification accuracy, whereas gestures requiring subtle muscle coordination, such as wrist pronation, exhibited slightly reduced performance. This may be because subtle elbow angle changes produce relatively small variations in muscle activation amplitude and recruitment patterns compared with differences between distinct gesture types. These findings suggest that the time-domain features used in this study effectively distinguish between gesture types but are less sensitive to the relatively small differences in muscle activation associated with the two tested angles. Therefore, incorporating frequency-domain or time–frequency features in future studies may improve sensitivity to subtle angle-related muscle activation changes.

Deep learning models, such as CNN- and RNN-based architectures, were excluded from this study. This is because they generally require large datasets, considerable computational resources, and long training times, although they often achieve high performance in sEMG-based classification. This study aimed to evaluate efficient, lightweight, and interpretable algorithms to initially investigate simultaneous gesture-and-angle classification; therefore, traditional machine learning models were considered more appropriate than deep learning ones. In addition, computational cost analysis demonstrated that the evaluated models maintained relatively low inference times and compact model sizes, supporting their potential applicability to lightweight HMI systems. As the dataset expands, future research may incorporate deep learning models to systematically compare performance and computational trade-offs.

Despite these promising results, several limitations should be acknowledged. First, although eight participants provided a sufficient number of trials for this preliminary

study, a cohort with increased size and diversity is required to improve generalizability across different ages, muscle anatomies, and physical conditions. Second, the experimental design included only four static gestures and two discrete elbow angles, intentionally selected to establish a controlled baseline. However, future studies should incorporate dynamic and continuous motions that better reflect real-world motor behavior. Third, feature extraction was restricted to time-domain features to maintain low computational cost. Although effective, this may offer limited sensitivity in angle discrimination. Incorporating frequency-domain or time–frequency features may provide additional information. Fourth, data were collected under controlled laboratory conditions that did not account for real-world factors, such as electrode displacement, sweat, long-term muscle fatigue, or daily variations in electrode placement. These factors can degrade signal quality and should be evaluated in future studies to ensure robustness. Finally, only traditional machine learning models were examined in this study. Comparing these models with deep learning approaches using the same dataset will clarify the accuracy-versus-efficiency trade-off.

Furthermore, the evaluation in this study was performed using randomized trial-wise validation, in which samples from the same participant could be included in both the training and testing sets. Therefore, the reported classification performance should be interpreted as performance within the collected dataset rather than subject-independent generalization to unseen individuals. Although per-subject classification accuracies were added to illustrate inter-individual variability, they were obtained under the trial-wise validation framework and therefore should not be interpreted as evidence of subject-independent generalization. Future studies should employ subject-independent validation strategies, such as Leave-One-Subject-Out (LOSO) cross-validation, using larger and more diverse participant cohorts.

In conclusion, this study demonstrates that traditional machine learning models can achieve high classification accuracy exceeding 95% in simultaneously classifying hand gestures and elbow angles using only sEMG signals. The findings validate the feasibility of sensor-efficient and computationally lightweight approaches for recognizing multi-dimensional motion intent. These results provide preliminary evidence for future studies on prosthetic control, rehabilitation robotics, and HMI systems. Future studies will expand participant diversity, incorporate dynamic and continuous movements, evaluate additional feature domains, explore deep learning-based models, and assess robustness under real-world conditions, such as electrode shifts, fatigue, and environmental variability.

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Author contributions Shinjeong Lee contributed to the conceptualization and design of the study, collected and analyzed the sEMG data, developed the algorithms and methodologies, implemented the machine learning models, and drafted the original manuscript. Junghun Kim contributed to the conceptualization of the study, supervised the overall research process, and critically reviewed and revised the manuscript. Sang-Il Choi contributed to the supervision of the study and participated in the review and editing of the manuscript. All authors read and approved the final manuscript.

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Data availability The raw sEMG data and extracted features that support the findings of this study are available from the corresponding author upon reasonable request, subject to participant-data and institutional policy restrictions.

Declarations

Conflict of interest The authors declare no conflict of interests.

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